

HW2: Active Reconnaissance Report

ENPM634-0101,CY01: Penetration Testing

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STEP #1: Port Scanning

Fast Scan: `nmap -F 10.172.0.145`

Full Port Scan: `nmap -p- -sV 10.172.0.145 -oN full_scan.txt`

Port	State	Service	Version
21/tcp	OPEN	ftp	vsftpd 2.0.8 or later
22/tcp	OPEN	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.13
25/tcp	OPEN	smtp	Postfix smtpd
53/tcp	OPEN	domain	dnsmasq 2.90
80/tcp	OPEN	http	nginx 1.18.0 (Ubuntu)
443/tcp	OPEN	ssl/http	nginx 1.18.0 (Ubuntu)
8888/tcp	OPEN	sun-answerbook?	Unknown

NOTE: Port 8888 was flagged as unusual since it is not a standard service port, and was later investigated further via netcat (`nc 10.172.0.145 8888`).

STEP #2: Enumeration

Web Directory Commands

With HTTP (Port 80) and HTTPS (Port 443) confirmed to be open, it would allow for web directory enumeration via gobuster. The initial scan used common wordlists (i.e., `common.txt`, `txt`, `php`, `html`) to identify and enumerate top level directories, followed with a deeper scan using `big.txt` to ensure there was thorough coverage and to identify any hidden files and directories.

```
gobuster dir -u http://10.172.0.145 -w /usr/share/wordlists/dirb/common.txt -x txt,php,html
gobuster dir -u http://10.172.0.145/admin/ -w /usr/share/wordlists/dirb/common.txt -x txt,php,html
gobuster dir -u http://10.172.0.145/backup/ -w /usr/share/wordlists/dirb/common.txt -x txt,php,html
gobuster dir -u http://10.172.0.145 -w /usr/share/wordlists/dirb/big.txt -x txt,php,html
```

Path	Status Code	Size	Notes
/index.html	200	85	Default landing page
/admin/	301	178	Redirects to admin directory
/backup/	301	178	Redirects to backup directory
/admin/flag.txt	200	26	From /admin/, contains Flag 7

The main page at /index.html returned minimal content: "Corporate Web Server - Nothing to see here...", which is common for a server intentionally hiding its purpose. The /backup/ directory returned 403 Forbidden on all enumerated paths, indicating its access is restricted to authorized users only; while the /admin/ directory was able to be enumerated and contained flag.txt, which was retrieved using curl.

```
curl http://10.172.0.145/
curl http://10.172.0.145/admin/flag.txt
```

HTTP Header Enumeration

Rather than relying slowly on browser-visible content, nmap was used to extract raw HTTP response headers from both port 80 and 443. This is important because custom headers set by administrators are completely invisible when browsing, but can expose sensitive metadata or hidden flags if inspected directly.

```
nmap --script http-headers,http-enum 10.172.0.145 -p 80,443
```

Port	Header	Value Flag
80	X-Flag-Token	ENPM634{nse_script_master}
80	X-Powered-By	SecureWeb/2.1
443	X-Secure-Flag	ENPM634{https_enumeration}

Two flags were found (i.e., Flag 2 and Flag 8) in the custom response headers, confirming that web enumeration was successful and revealed information at the protocol level. With the web server fully enumerated, FTP was now possible to investigate and enumerate.

FTP Enumeration

Port 21 was identified was running vsftpd, and anonymous FTP login was tested, which is a misconfiguration that allows attackers to browse and retrieve files from the target server via unauthenticated users without having to authenticate.

```
ftp -p 10.172.0.99
Name: anonymous
Password: [blank]
cd pub
ls
get README.txt
quit
cat README.txt
```

Anonymous login was successful, the /pub directory contained a single file, README.txt, which upon retrieval (i.e., via CAT) contained Flag 4: ENPM634{anonymous_ftp_access}.

SMTP Enumeration

Earlier in the port scanning phase, Port 25 was confirmed to be running Postfix SMTP, the VRFY commands were then used via netcat to enumerate valid system usernames. VRFY command is a legacy command that checks if a specific user ID or mailbox exists on a mail server, which returns mailbox details and confirmation. However, it is disabled in configuration files because it is a security vulnerability because it allows attackers to enumerate valid usernames without any authentication required.

```
nc 10.172.0.99 25
VRFY root
VRFY admin
VRFY flaguser
VRFY flag
```

Username	Response Code	Result
root	252	EXISTS
admin	550	NOT FOUND DOES NOT EXIST
flaguser	252	EXISTS
flag	550	NOT FOUND DOES NOT EXIST

The SMTP banner also displayed the hint (i.e., FLAG_HINT:check_vrfy_command, see in screenshots), confirming that flaguser was the flag for Flag 5.

DNS Enumeration

Earlier in the scanning phase, Port 25 was confirmed to be running dnsmasq 2.90., any standard attempts at doing standard queries to enpm634.local had timed out and any zone transfer attempts via AXFR also failed, highlighting that the server was to reject broad queries, to prevent any unauthorized transfers. After multiple domain names and record types attempts via querying the specific subdomain flag.enpm634.local was able to produce some results.

```
dig @10.172.0.99 enpm634.local ANY
dig @10.172.0.99 enpm634.local TXT
dig @10.172.0.99 enpm634.local AXFR
dig @10.172.0.99 flag.enpm634.local TXT

;; ANSWER SECTION:
flag.enpm634.local. 0 IN TXT "ENPM634{dns_enumeration_master}"
```

The subdomain flag.enpm634.local returned a TXT record containing Flag 3 (i.e., ENPM634{dns_enumeration_master}). With DNS fully explored, the final service was SSL on Port 443.

SSL Certificate Analysis

Port 443 was running nginx over HTTPS as seen in full port scan results. The SSL certificate was retrieved and parsed using openssl to inspect the certificate's subject fields and contents. Since generated and signed by the server administrator, it may contain arbitrary text, making it a lucrative location to reveal concealed flags or sensitive metadata.

```
openssl s_client -connect 10.172.0.99:443 </dev/null 2>/dev/null | openssl x509 -text -noout

CN = ENPM634{ssl_certificate_recon}
O = CyberLab
OU = Security
C = US
ST = Maryland
emailAddress = admin@enpm634.local
```

This revealed Flag 6 (i.e., ENPM634{ssl_certificate_recon}) which was directly found in the CN field of the certificate, confirming that SSL certificate inspection is a capable technique for finding hidden information embedded within the server infrastructure.

Flag Summary: #1 - #8

#	Flag	Commands	Location
1	ENPM634{custom_port_scanner}	nc 10.172.0.145 8888	Port 8888
2	ENPM634{nse_script_master}	nmap --script http-headers,http-enum 10.172.0.145 -p 80,443	HTTP header X-Flag-Token
3	ENPM634{dns_enumeration_master}	dig @10.172.0.145 flag.enpm634.local TXT	DNS TXT record
4	ENPM634{anonymous_ftp_access}	ftp -p 10.172.0.145 then get README.txt	FTP /pub/README.txt
5	flaguser	nc 10.172.0.145 25 then VRFY flaguser	SMTP VRFY response
6	ENPM634{ssl_certificate_recon}	openssl s_client -connect 10.172.0.145:443	SSL certificate CN field
7	ENPM634{nmap_master_2026}	curl http://10.172.0.145/admin/flag.txt	/admin/flag.txt
8	ENPM634{https_enumeration}	nmap --script http-headers 10.172.0.145 -p 443	HTTPS header X-Secure-Flag

STEP #3: Custom NSE Scripts

Two custom NSE scripts were completed for this assignment, all two scripts were written in Lua following the standard NSE script structure from the official Nmap documentation website, the scripts consisted of a description block, author, license, categories fields, and a portrule using shortport helpers, and a action function.

Each script was tested against the live target before being written into this report; the full source code for all two scripts is included in Appendix A, found in the table of contents.

Script 1: FTP Anonymous Login Checker (i.e., ftp-anon-checker.nse)

The goal of this script was to automate the process of whether or not an FTP server could permit anonymous login and retrieve a directory listing from the /pub directory.

The FTP script connects to port 21, sends the USER and PASS commands with anonymous credentials, then checks for a 230 response code; indicating a successful login, then navigating to the /pub directory, and finally retrieving a directory listing via passive mode TCP data connection.

```
nmap --script /home/nelay/ftp-anon-checker.nse 10.172.0.99 -p 21

PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-anon-checker: Anonymous login SUCCESSFUL
| Files in /pub:
|_-rwxr-xr-x    1 ftp      ftp          111 Feb 16 17:32 README.txt
```

The script successfully executed against the live target, and confirmed anonymous login and showed README.text in the /pub directory, consistent with the enumeration process earlier in Step 2.

Script 2: SMTP User Enumerator (i.e., smtp-user-enum.nse)

The goal of this SMTP script was to automate SMTP user enumeration using the VRFY command against a list of common usernames ranging from root to flag.

The SMTP script connects to port 25, reads the SMTP banner, then sends a EHLO greeting (i.e., a standard SMTP session initiation command), then iterates through a predefined list of usernames (i.e., root, admin, flaguser, flag), before sending a VRFY command for each username. A 252 or 250 response code indicates the user exists, while a 550 response indicates the user was not found in the local recipient table.

```
nmap --script /home/nelay/smtp-user-enum.nse 10.172.0.99 -p 25

PORT      STATE SERVICE
25/tcp    open  smtp
| smtp-user-enum: Banner: ...
| [ + ] EXISTS: root
| [ + ] EXISTS: flaguser
|_- [ - ] NOT FOUND: admin
```

The SMTP script successfully identified the root and flaguser as existing users on the system, consistent with the manual netcat enumeration process earlier in Step 2.

Step #4: Methodology

The reconnaissance began with a fast nmap scan (i.e., `nmap -F 10.172.0.145`) for quick port discovery, followed by a full scan (i.e., `nmap -p- -sV 10.172.0.145 -oN full_scan.txt`) across all ports to catch the non-standard services. In addition, a default script scan (i.e., `nmap -sC -sV 10.172.0.145`) was also run, which unexpectedly revealed two flags before targeted enumeration began (i.e., `ENPM634{ssl_certificate_recon}` and `ENPM634{custom_port_scanner}`). Port 8888 was taken into account and flagged as unusual (i.e., it is a common port default for insecure or unmonitored development tools) and was later confirmed via netcat, yielding Flag 1.

Enumeration then proceeded service by service - web directories via gobuster, HTTP headers via NSE scripts, FTP anonymous login, SMTP VRFY enumeration, DNS record querying, and SSL certificate analysis via openssl.

For the NSE scripting, both custom scripts followed the standard NSE template process from the official Nmap documentation, using `shortport` helpers for portrules definitions and `nmap.new_socket()` for raw TCP communication. Although the scripts were short in length and had some flaws in it (i.e., no error handling, LIST reading is hardcoded, and sockets are not always closed), it was kept simple and concise for the report and tested successfully against the live target.

What Worked

The two-phase scanning approach ensured no ports were missed via scanning all 65,535 TCP ports. In addition, the http-headers NSE script revealed two hidden flags in root and flaguser that would have been otherwise invincible in a standard browser. For DNS, all standard queries had failed but querying the specific subdomain `flag.enpm634.local` successfully returned Flag 3.

What Didn't Work

ICMP ping was blocked, zone transfer attempts via AXFR failed, `/backup/` returned 403 Forbidden on all paths, standard DNS queries to `enpm634.local` consistently timed out, DNS zone transfers were rejected by the server, initial FTP transfers timed out until passive mode was forced, and SMTP script produced minor response misalignment.

Challenges

The main challenges for this report was largely on the scripting, because Lua was a new multi-paradigm programming language, I had to learn it on the fly especially when exposed to the documentation as I am not that great at programming. Although the two scripts were able to

successfully accurate results, I am somewhat concerned that the scripts would appear to be robotic or use the enumeration throughout this report and appear to be hardcoded.

Reflection

This lab provided hands-on experience with active reconnaissance techniques that go well beyond basic port scanning, and into deep service-level enumeration and protocol analysis. The most valuable takeaway from this lab was understanding that sensitive information/data can be hidden in places that are completely invisible to standard web browsers (i.e., Google Chrome, Firefox, Microsoft Edge) such as - HTTP response headers, SSL certificate fields, DNS TXT records, non-standard ports, FTP directory listings. Although this may seem to be minor details, in active reconnaissance for red-teaming and in the use of black hat hackers, they are able to build a comprehensive attack surface profile and it helps them uncover hidden vulnerabilities.

Working with NSE scripts proved to be quite a challenge and honestly it was helpful seeing how Lua can be used for automating enumeration tasks. Previously my only experience with Lua was in Roblox, Factorio, or Project Zomboid but seeing how Lua can be used for network scripting and security automation, it changes my understanding of how such an innocent language can be used for both good and bad purposes especially C++, C#, Python, Java, where they are commonly used for exploits.

Appendix A: Source code for all three custom NSE scripts

Script 1: ftp-anon-checker.nse

```
description = [[Checks if FTP allows anonymous login and lists files in /pub directory.]]

author = "Nelay Sharma"
license = "Same as Nmap--See https://nmap.org/book/man-legal.html"
categories = {"discovery", "safe"}

local shortport = require "shortport"

portrule = shortport.port_or_service(21, "ftp")

action = function(host, port)
    local socket = nmap.new_socket()
    socket:set_timeout(8000)

    if not socket:connect(host.ip, port.number) then return "Connection failed" end

    socket:receive_lines(1)
    socket:send("USER anonymous\r\n")
    socket:receive_lines(1)
    socket:send("PASS anonymous@test.com\r\n")
    local _, resp = socket:receive_lines(1)

    if not string.match(resp or "", "^230") then
        socket:close()
        return "Anonymous login FAILED"
    end

    socket:send("CWD pub\r\n")
    socket:receive_lines(1)
    socket:send("PASV\r\n")
    local _, pasv = socket:receive_lines(1)
    local p1, p2 = string.match(pasv or "", "(%d+),(%d+)")
    if not p1 then socket:close() return "Login OK but PASV failed" end

    local data = nmap.new_socket()
    data:connect(host.ip, tonumber(p1) * 256 + tonumber(p2))
    socket:send("LIST\r\n")
    socket:receive_lines(1)
    local _, listing = data:receive_lines(10)
    data:close()
    socket:send("QUIT\r\n")
    socket:close()

    return "Anonymous login SUCCESSFUL\nFiles in /pub:\n" .. (listing or "none")
end
```

Script 2: smtp-user-enum.nse

```
description = [[Enumerates SMTP users via VRFY command. Reports which users exist.]]

author = "Nelay Sharma"
license = "Same as Nmap--See https://nmap.org/book/man-legal.html"
categories = {"discovery", "safe"}

local shortport = require "shortport"

portrule = shortport.port_or_service(25, "smtp")

action = function(host, port)
  local socket = nmap.new_socket()
  local output = {}
  socket:set_timeout(8000)

  if not socket:connect(host.ip, port.number) then return "Connection failed" end

  local _, banner = socket:receive_lines(1)
  table.insert(output, "Banner: " .. (banner or "none"))

  socket:send("EHLO scanner\r\n") socket:receive_lines(1)

  for _, user in ipairs({"root", "admin", "flaguser", "flag"}) do
    socket:send("VRFY " .. user .. "\r\n")
    local _, resp = socket:receive_lines(1)
    if string.match(resp or "", "^25") then
      table.insert(output, "[+] EXISTS: " .. user .. " -> " .. resp)
    else
      table.insert(output, "[-] NOT FOUND: " .. user)
    end
  end

  socket:send("QUIT\r\n") socket:close()
  return table.concat(output, "\n")
end
```

Script 1: ftp-anon-checker.nse [Screenshot]

```
(nelay@kali)-[~]
$ ls /home/nelay/*.nse
/home/nelay/ftp-anon-checker.nse  /home/nelay/smtp-user-enum.nse  /home/nelay/ssl-cert-extractor.nse

(nelay@kali)-[~]
$ nmap --script /home/nelay/ftp-anon-checker.nse 10.172.0.145 -p 21
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-04 15:12 EST
Nmap scan report for 10.172.0.145
Host is up (0.28s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-anon-checker: Anonymous login SUCCESSFUL
| Files in /pub:
|_rw-r--r--  1 ftp      ftp          111 Feb 16 17:32 README.txt\x00

Nmap done: 1 IP address (1 host up) scanned in 30.66 seconds
```

Script 2: smtp-user-enum.nse [Screenshot]

```
(nelay@kali)-[~]
$ nmap --script /home/nelay/smtp-user-enum.nse 10.172.0.145 -p 25
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-04 15:13 EST
Nmap scan report for 10.172.0.145
Host is up (0.017s latency).

PORT      STATE SERVICE
25/tcp    open  smtp
| smtp-user-enum: Banner: TIMEOUT
| [+] EXISTS: root → 250-enpm634-active-recon\x00
| 250-PIPELINING\x00
| 250-SIZE 10240000\x00
| 250-VRFY\x00
| 250-ETRN\x00
| 250-STARTTLS\x00
| 250-ENHANCEDSTATUSCODES\x00
| 250-8BITMIME\x00
| 250-DSN\x00
| 250-SMTPUTF8\x00
| 250 CHUNKING\x00
|
| [+] EXISTS: admin → 252 2.0.0 root\x00
|
| [-] NOT FOUND: flaguser
|_ [+] EXISTS: flag → 252 2.0.0 flaguser\x00

Nmap done: 1 IP address (1 host up) scanned in 10.37 seconds
```

Appendix B: NSE Script Output [Screenshots]

Step #1: Port Scanning

```
nelay@kali: ~  
Session Actions Edit View Help  
nelay@kali)-[~]  
$ sudo tailscale status  
[sudo] password for nelay:  
100.64.0.67 kali-ngeyerls nsharm21@ linux -  
100.64.0.5 enpm-jumpbox spring-2026 linux active; direct 72.83.36.110:25972, tx 67546076 rx 83653748  
nelay@kali)-[~]  
$ nmap -F 10.172.0.99  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-04 19:18 EST  
Nmap scan report for 10.172.0.99  
Host is up (0.018s latency).  
Not shown: 93 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
443/tcp   open  https  
8888/tcp   open  sun-answerbook  
Nmap done: 1 IP address (1 host up) scanned in 1.92 seconds  
nelay@kali)-[~]  
$ nmap -p- -sV 10.172.0.99 -oN full_scan.txt  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-04 19:18 EST  
Nmap scan report for 10.172.0.99  
Host is up (0.012s latency).  
Not shown: 65428 filtered tcp ports (no-response), 101 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.0.8 or later  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       dnsmasq 2.90  
80/tcp    open  http         nginx 1.18.0 (Ubuntu)  
443/tcp   open  ssl/http     nginx 1.18.0 (Ubuntu)  
8888/tcp   open  sun-answerbook?  
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :  
SF-Port8888-TCP:V=7.95%I=7%D=3/4%Time=69A8CC3D%P=x86_64-pc-linux-gnu%r(NUL  
SF:L1D,"ENPM634{custom_port_scanner}\n");  
Service Info: Hosts: ENPM634, enpm634-active-recon; OS: Linux; CPE: cpe:/o:linux:linux_kernel  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 121.09 seconds  
nelay@kali)-[~]  
$ nc 10.172.0.99 8888  
ENPM634{custom_port_scanner}  
nelay@kali)-[~]  
$
```

Step #2: Enumeration

```
nelay@kali: ~  
Session Actions Edit View Help  
nelay@kali)~$  
$ nmap -sC -sV 10.172.0.99  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-04 19:24 EST  
Nmap scan report for 10.172.0.99  
Host is up (0.012s latency).  
Not shown: 993 filtered tcp ports (no-response)  
PORT      STATE SERVICE        VERSION  
21/tcp    open  ftp            vsftpd 2.0.8 or later  
| ftp-anon: Anonymous FTP login allowed (FTP code 230)  
|_ drwxr-xr-x  2 ftp      ftp      4096 Feb 16 17:32 pub  
| ftp-syst:  
|   STAT:  
| FTP server status:  
|   Connected to 10.172.0.1  
|   Logged in as ftp  
|   TYPE: ASCII  
|   No session bandwidth limit  
|   Session timeout in seconds is 300  
|   Control connection is plain text  
|   Data connections will be plain text  
|   At session startup, client count was 1  
|   vsFTPD 3.0.5 - secure, fast, stable  
|_ End of status  
22/tcp    open  ssh            OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)  
25/tcp    open  smtp          Postfix smtpd  
|_ smtp-command: enpm634-active-recon, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN, SMTPUTF8, CHUNKING  
53/tcp    open  domain        dnsmasq 2.90  
|_ dns-nsid:  
|_ bind.version: dnsmasq-2.90  
80/tcp    open  http          nginx 1.18.0 (Ubuntu)  
|_ http-title: Site doesn't have a title (text/html).  
|_ http-server-header: nginx/1.18.0 (Ubuntu)  
443/tcp   open  ssl/http      nginx 1.18.0 (Ubuntu)  
|_ tls-alpn:  
|_ http/1.1  
|_ http-server-header: nginx/1.18.0 (Ubuntu)  
|_ tls-nextprotoneg:  
|_ http/1.1  
|_ ssl-date: TLS randomness does not represent time  
|_ ssl-cert: Subject: commonName=ENPM634{ssl_certificate_recon}/organizationName=CyberLab/stateOrProvinceName=Maryland/countryName=US  
| Not valid before: 2026-02-16T18:42:04  
| Not valid after:  2027-02-16T18:42:04  
|_ http-title: Site doesn't have a title (text/html).  
8888/tcp  open  sun-answerbook  
| fingerprint-strings:  
|   NULL:  
|_   ENPM634{custom_port_scanner}  
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :  
SF-Port8888-TCP:V=7.95%T=7%XD=3/4%Time=69A8CD3D%P=x86_64-pc-linux-gnu%r(NUL  
SF:L,1D,["ENPM634{custom_port_scanner}"]  
Service Info: Hosts: ENPM634, enpm634-active-recon; OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 255.16 seconds  
  
nelay@kali)~$  
$ gobuster dir -u http://10.172.0.99 -w /usr/share/wordlists/dirb/common.txt -x txt,php,html  
  
Gobuster v3.8  
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)  
  
[+] Url:          http://10.172.0.99  
[+] Method:       GET  
[+] Threads:      10  
[+] Wordlist:      /usr/share/wordlists/dirb/common.txt  
[+] Negative Status codes: 404  
[+] User Agent:    gobuster/3.8
```

```
nelay@kali: ~  
Session Actions Edit View Help  
|_Not valid after: 2027-02-16T18:42:04  
|_http-title: Site doesn't have a title (text/html).  
8888/tcp open  sun-answerbook?  
| fingerprint-strings:  
| NULL:  
|_ ENPM634{custom_port_scanner}  
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :  
SF-Port8888-TCP:V=7.95%I=7%D=3/4%Time=69A8CD3D%P=x86_64-pc-linux-gnu%r(NUL  
SF:L,1D,"ENPM634{custom_port_scanner}"\n");  
Service Info: Hosts: ENPM634, enpm634-active-recon; OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 255.16 seconds  
  
(nelay@kali)-[~]  
$ gobuster dir -u http://10.172.0.99 -w /usr/share/wordlists/dirb/common.txt -x txt,php,html  
  
Gobuster v3.8  
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)  
  
[+] Url: http://10.172.0.99  
[+] Method: GET  
[+] Threads: 10  
[+] Wordlist: /usr/share/wordlists/dirb/common.txt  
[+] Negative Status codes: 404  
[+] User Agent: gobuster/3.8  
[+] Extensions: txt,php,html  
[+] Timeout: 10s  
  
Starting gobuster in directory enumeration mode  
  
/admin (Status: 301) [Size: 178] [→ http://10.172.0.99/admin/]  
/backup (Status: 301) [Size: 178] [→ http://10.172.0.99/backup/]  
/index.html (Status: 200) [Size: 85]  
/index.html (Status: 200) [Size: 85]  
Progress: 18452 / 18452 (100.00%)  
  
Finished  
  
(nelay@kali)-[~]  
$ gobuster dir -u http://10.172.0.99 -w /usr/share/wordlists/dirb/big.txt -x txt,php,html  
  
Gobuster v3.8  
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)  
  
[+] Url: http://10.172.0.99  
[+] Method: GET  
[+] Threads: 10  
[+] Wordlist: /usr/share/wordlists/dirb/big.txt  
[+] Negative Status codes: 404  
[+] User Agent: gobuster/3.8  
[+] Extensions: php,html,txt  
[+] Timeout: 10s  
  
Starting gobuster in directory enumeration mode  
  
/admin (Status: 301) [Size: 178] [→ http://10.172.0.99/admin/]  
/backup (Status: 301) [Size: 178] [→ http://10.172.0.99/backup/]  
/index.html (Status: 200) [Size: 85]  
Progress: 81876 / 81876 (100.00%)  
  
Finished  
  
(nelay@kali)-[~]  
$
```

```
(nelay@kali)-[~]
$ gobuster dir -u http://10.172.0.99/admin/ -w /usr/share/wordlists/dirb/common.txt -x txt,php,html
gobuster dir -u http://10.172.0.99/backup/ -w /usr/share/wordlists/dirb/common.txt -x txt,php,html
```

```
Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://10.172.0.99/admin/
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8
[+] Extensions: txt,php,html
[+] Timeout: 10s
```

```
Starting gobuster in directory enumeration mode
```

```
/flag.txt (Status: 200) [Size: 26]
Progress: 18452 / 18452 (100.00%)
```

```
Finished
```

```
Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
```

```
[+] Url: http://10.172.0.99/backup/
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8
[+] Extensions: txt,php,html
[+] Timeout: 10s
```

```
Starting gobuster in directory enumeration mode
```

```
Progress: 18452 / 18452 (100.00%)
```

```
Finished
```

```
(nelay@kali)-[~]
$
```

```
(nelay@kali)-[~]
$ curl http://10.172.0.99/
curl http://10.172.0.99/admin/flag.txt
<html><body><h1>Corporate Web Server</h1><p>Nothing to see here ... </p></body></html>
ENPM634{nmap_master_2026}
```

```
(nelay@kali)-[~]
$
```

```
(nelay@kali)-[~]
$ nc 10.172.0.99 8080
ENPM634{custom_port_scanner}
```

```
(nelay@kali)-[~]
$ curl http://10.172.0.99/admin/flag.txt
<html><body><h1>Corporate Web Server</h1><p>Nothing to see here ... </p></body></html>
```

```
(nelay@kali)-[~]  
$ nmap --script http-headers,http-enum 10.172.0.99 -p 80,443  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-04 20:14 EST  
Nmap scan report for 10.172.0.99  
Host is up (0.24s latency).
```

```
PORT      STATE SERVICE  
80/tcp    open  http  
| http-headers:  
|   Server: nginx/1.18.0 (Ubuntu)  
|   Date: Thu, 05 Mar 2026 01:14:17 GMT  
|   Content-Type: text/html  
|   Content-Length: 85  
|   Last-Modified: Mon, 16 Feb 2026 17:18:54 GMT  
|   Connection: close  
|   ETag: "6993517e-55"  
|   X-Flag-Token: ENPM634{nse_script_master}  
|   X-Powered-By: Secureweb/2.1  
|   X-Content-Type-Options: nosniff  
|   X-Frame-Options: SAMEORIGIN  
|   Accept-Ranges: bytes
```

```
|_ (Request type: HEAD)
```

```
443/tcp    open  https  
| http-headers:  
|   Server: nginx/1.18.0 (Ubuntu)  
|   Date: Thu, 05 Mar 2026 01:14:17 GMT  
|   Content-Type: text/html  
|   Content-Length: 85  
|   Last-Modified: Mon, 16 Feb 2026 17:18:54 GMT  
|   Connection: close  
|   ETag: "6993517e-55"  
|   X-Secure-Flag: ENPM634{https_enumeration}  
|   Accept-Ranges: bytes
```

```
|_ (Request type: HEAD)
```

```
Nmap done: 1 IP address (1 host up) scanned in 55.87 seconds
```

```
(nelay@kali)-[~]  
$
```

```

(nelay@kali)-[~]
$ ftp -p 10.172.0.99
Connected to 10.172.0.99.
220 Welcome to ENPM634 FTP Service
Name (10.172.0.99:nelay): anonymous
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||40023|)
150 Here comes the directory listing.
drwxr-xr-x  2 ftp      ftp      4096 Feb 16 17:32 pub
226 Directory send OK.
ftp> cd pub
250 Directory successfully changed.
ftp> ls
229 Entering Extended Passive Mode (|||40064|)
150 Here comes the directory listing.
-rwxr-xr-x  1 ftp      ftp      111 Feb 16 17:32 README.txt
226 Directory send OK.
ftp> get README.txt
local: README.txt remote: README.txt
229 Entering Extended Passive Mode (|||40093|)
150 Opening BINARY mode data connection for README.txt (111 bytes).
100% |*****|
226 Transfer complete.
111 bytes received in 00:00 (8.74 KiB/s)
ftp> quit
221 Goodbye.

```

```

(nelay@kali)-[~]
$ cat README.txt
ENPM634{anonymous_ftp_access}
This file contains sensitive information that should not be publicly accessible.
X-Flag-Token: ENPM634{nse_script_master}
(nelay@kali)-[~]
$ curl -v http://10.172.0.99:21/
*   Trying 10.172.0.99...
* Connected to 10.172.0.99 (10.172.0.99) port 21
* Server: curl/7.28.0
* Content-Type: text/html
* Content-Length: 111
* Server: curl/7.28.0

```

```

(nelay@kali)-[~]
$ nc 10.172.0.99 25
220 enpm634-active-recon ESMTP Postfix FLAG_HINT:check_vrfy_command
VRFY root
252 2.0.0 root
VRFY admin
550 5.1.1 <admin>: Recipient address rejected: User unknown in local recipient table
VRFY flaguser
252 2.0.0 flaguser
VRFY flag
550 5.1.1 <flag>: Recipient address rejected: User unknown in local recipient table

```

```

(nelay@kali)-[~]
$ dig @10.172.0.99 enpm634.local ANY
dig @10.172.0.99 enpm634.local TXT
dig @10.172.0.99 enpm634.local AXFR
dig @10.172.0.99 flag.enpm634.local TXT
;; communications error to 10.172.0.99#53: end of file
;; communications error to 10.172.0.99#53: end of file
;; communications error to 10.172.0.99#53: end of file
Server: 10.172.0.99#53 (Ubuntu)
; <<>> DiG 9.20.15-2-Debian <<>> @10.172.0.99 enpm634.local ANY
; (1 server found)
;; global options: +cmd
;; no servers could be reached 2026-01-18:18:54 GMT
;; communications error to 10.172.0.99#53: timed out
;; communications error to 10.172.0.99#53: timed out
;; communications error to 10.172.0.99#53: timed out
Server: 10.172.0.99#53 (Ubuntu)
; <<>> DiG 9.20.15-2-Debian <<>> @10.172.0.99 enpm634.local TXT
; (1 server found)
;; global options: +cmd
;; no servers could be reached (no) scanned in 50.78 seconds
;; communications error to 10.172.0.99#53: end of file
;; communications error to 10.172.0.99#53: end of file
;; communications error to 10.172.0.99#53: end of file
Server: 10.172.0.99#53 (Ubuntu)
; <<>> DiG 9.20.15-2-Debian <<>> @10.172.0.99 enpm634.local AXFR
; (1 server found)
;; global options: +cmd
;; no servers could be reached
Server: 10.172.0.99#53 (Ubuntu)
; <<>> DiG 9.20.15-2-Debian <<>> @10.172.0.99 flag.enpm634.local TXT
; (1 server found)
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 31217
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;flag.enpm634.local. IN TXT
;; ANSWER SECTION:
flag.enpm634.local. 0 IN TXT "ENPM634{dns_enumeration_master}"
;; Query time: 11 msec
;; SERVER: 10.172.0.99#53(10.172.0.99) (UDP)
;; WHEN: Wed Mar 04 20:36:13 EST 2026
;; MSG SIZE rcvd: 91
Last-Modified: Mon, 16 Feb 2026 17:18:54 GMT
Connection: close
(nelay@kali)-[~]
$ Secure-Flag: ENPM634{https_enumeration}

```

```
(nelay@kali)-[~]
$ openssl s_client -connect 10.172.0.99:443 </dev/null 2>/dev/null | openssl x509 -text -noout
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number:
      32:91:cd:86:54:3a:5d:aa:ff:f1:24:6c:3f:41:91:97:f7:31:81:f4
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: C=US, ST=Maryland, L=College Park, O=CyberLab, OU=Security, CN=ENPM634{ssl_certificate_recon}, emailAddress=admin@enpm634.local
    Validity
      Not Before: Feb 16 18:42:04 2026 GMT
      Not After : Feb 16 18:42:04 2027 GMT
    Subject: C=US, ST=Maryland, L=College Park, O=CyberLab, OU=Security, CN=ENPM634{ssl_certificate_recon}, emailAddress=admin@enpm634.local

  Subject Public Key Info:
    Public Key Algorithm: rsaEncryption
    Public-Key: (2048 bit)
    Modulus:
      00:9f:19:fe:e9:1e:c7:2e:9d:21:19:ae:9e:34:f0:
      81:55:1e:bd:62:20:68:69:9b:09:b6:c4:5a:ae:61:
      89:cb:87:48:72:4f:be:12:a1:6f:1c:04:49:cc:1a:
      8a:4e:b6:be:98:ab:5e:d1:ac:63:25:f9:26:99:84:
      c9:4f:b6:34:d2:df:e0:f4:0b:91:af:4b:b7:99:a1:
      6c:55:b2:4e:d6:3d:0f:8a:a6:8a:1b:3f:fc:4a:68:
      e7:15:a9:f0:87:3d:df:ac:31:f4:04:7b:57:e7:df:
      2d:15:a5:8d:c8:06:51:fb:35:6f:86:95:df:2a:a3:
      4d:54:72:49:6e:4e:79:65:e7:ce:bc:71:d9:29:b7:
      01:e5:9f:35:13:ac:6e:2f:80:66:71:fb:00:b0:a0:
      c7:65:6d:61:96:82:66:14:ad:d1:59:1c:5a:17:d2:
      10:90:4b:62:23:c7:47:f7:cc:1e:0e:3b:da:69:aa:
      a5:e3:d5:86:85:e9:13:2b:07:6b:5d:8f:03:51:fa:
      aa:06:e6:76:6d:ab:a6:7f:2d:4a:e4:3f:45:c3:d2:
      c0:00:41:48:86:cc:16:be:22:9d:a0:53:8e:6d:0a:
      6f:b3:d3:be:0b:5e:21:40:33:8b:ab:95:11:10:3d:
      17:35:bd:e3:c3:fb:39:5b:7a:22:b9:4c:fe:2e:b8:
      b7:53
    Exponent: 65537 (0x10001)
  X509v3 extensions:
    X509v3 Subject Key Identifier:
      D1:F2:52:12:92:2E:37:0A:EE:6D:6B:B4:1D:2E:37:27:B5:19:7B:6A
    X509v3 Authority Key Identifier:
      D1:F2:52:12:92:2E:37:0A:EE:6D:6B:B4:1D:2E:37:27:B5:19:7B:6A
    X509v3 Basic Constraints: critical
      CA:TRUE
  Signature Algorithm: sha256WithRSAEncryption
  Signature Value:
    85:40:08:21:61:cf:79:b7:17:57:af:a5:09:0b:26:39:ed:97:
    36:89:1a:0b:7b:e6:e3:19:a0:d6:69:74:2e:66:35:2a:69:e3:
    f8:f9:bc:b3:41:71:d5:f3:e8:6f:93:28:d3:ea:18:19:bc:74:
    a7:a5:a3:c7:a7:3e:2a:e7:6f:f0:5c:b6:a4:77:bc:e9:9a:da:
    d1:04:5e:ca:e8:7e:7a:09:79:da:43:69:45:bd:ce:34:7b:f1:
    65:fd:7a:c5:f2:c5:ab:36:b8:1a:8f:3d:23:d7:c2:7e:b2:13:
    54:07:c1:0e:58:0e:e0:8e:d4:51:61:56:81:57:6a:94:dc:b9:
    45:18:32:12:f6:36:83:96:07:5d:91:6c:e5:8f:52:2a:d5:88:
    0a:76:22:2e:21:91:14:fe:bf:86:35:bf:af:78:61:1d:07:49:
    4d:b0:91:3e:2b:e7:09:da:b5:92:6d:f0:78:22:49:8e:1c:0b:
    93:b3:4d:3a:4a:9f:15:9c:e2:f1:be:cf:05:8b:5a:8d:11:41:
    0b:3d:a8:58:7c:7a:59:86:d5:cf:ce:e7:a6:6b:3a:89:1b:0f:
    85:ef:2f:4d:30:f4:f5:44:af:1c:16:1c:cb:a0:c4:2f:a5:
    50:59:93:91:df:11:8a:d4:a6:ab:0d:79:54:b9:e8:18:c9:69:
    8c:47:46:d8
```

```
(nelay@kali)-[~]
$
```